

Zero-Shot Prompt

You are a helpful assistant for analyzing data visualizations.
 Answer the following multiple-choice question using ONLY information visible in the chart image.
 Do not use outside knowledge.

Question: {question}
 Options: {formatted_options}

Choose exactly one option from ({opt_range}). Respond with nothing else besides the following line, replacing <number> with your choice:
 ANSWER: <number>

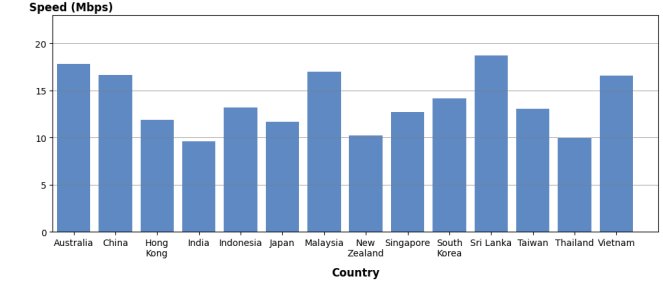
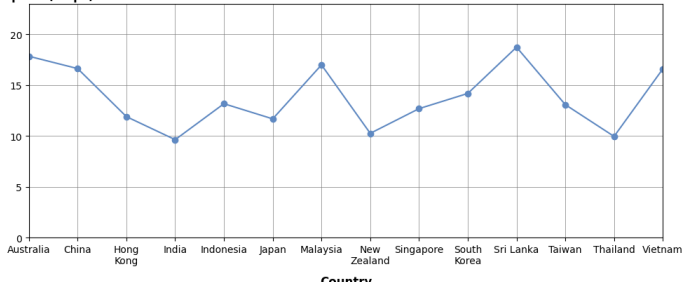
Figure 5: Prompt used for the zero-shot task.

A Appendix: Prompt Templates

Figure 5 shows the prompt template used for zero-shot chart question answering, where question, formatted_options, and opt_range are populated from the corresponding benchmark item.

B Appendix: Charts

This appendix collects the visual representations of the six datasets used in the study. For each chart, the missing counterpart chart type was produced from the same underlying data, allowing us to compare how question difficulty and interpretation vary across encodings. The table below lists each rendered image alongside its chart number, variable type (univariate or multivariate), and plot style.

Image	Chart
Average Internet Speeds in Asia  Speed (Mbps) Australia China Hong Kong India Indonesia Japan Malaysia New Zealand Singapore South Korea Sri Lanka Taiwan Thailand Vietnam Country	Chart 1 - Univariate - Bar plot
Average Internet Speeds in Asia  Speed (Mbps) Australia China Hong Kong India Indonesia Japan Malaysia New Zealand Singapore South Korea Sri Lanka Taiwan Thailand Vietnam Country	Chart 1 - Univariate - Line plot

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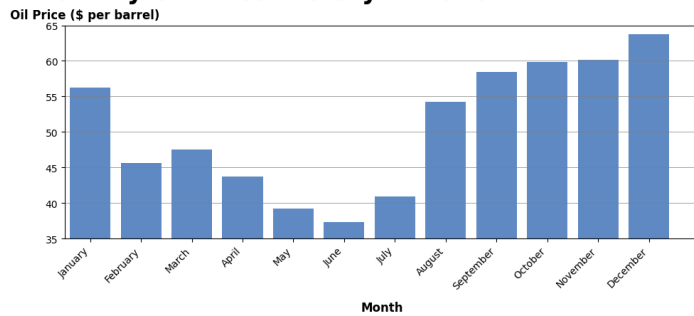
Monthly Oil Price History in 2015

Chart 2 - Univariate - Bar plot

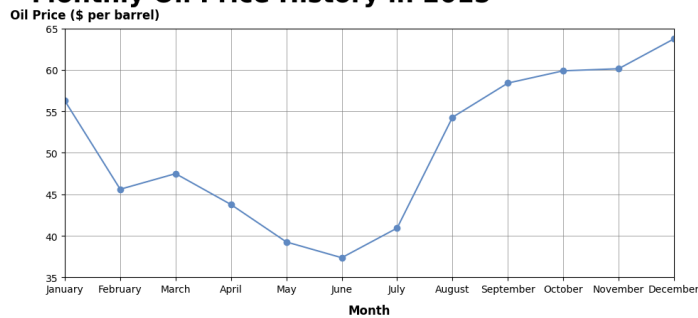
Monthly Oil Price History in 2015

Chart 2 - Univariate - Line plot

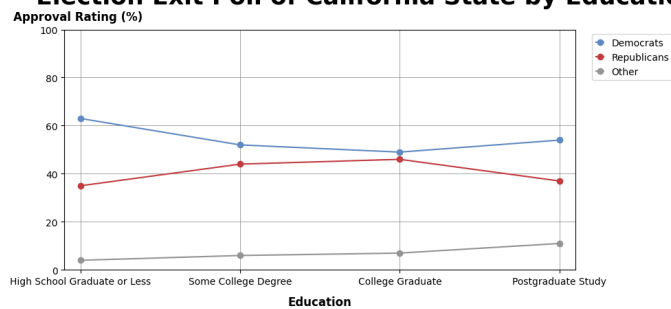
Election Exit Poll of California State by Education

Chart 3 - Multivariate - Line plot

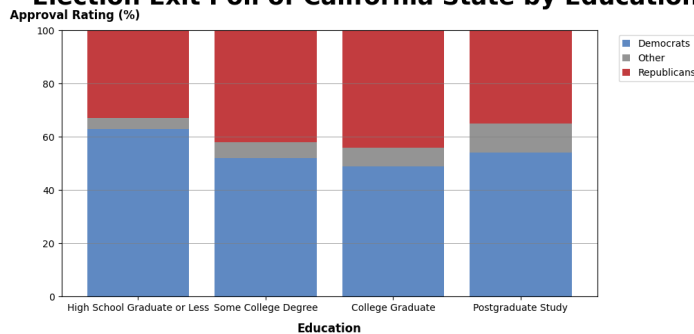
Election Exit Poll of California State by Education

Chart 3 - Multivariate - Stacked Bar plot

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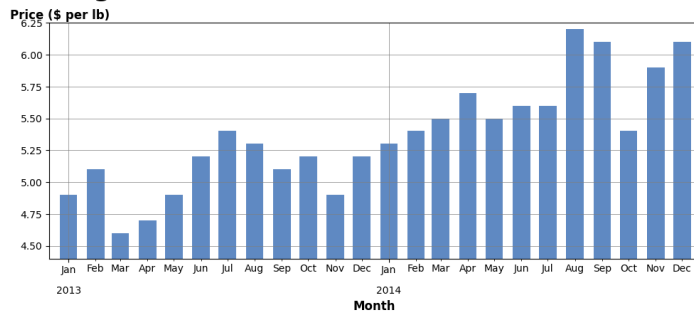
Average Coffee Bean Price from 2013 to 2014

Chart 4 - Univariate - Bar plot

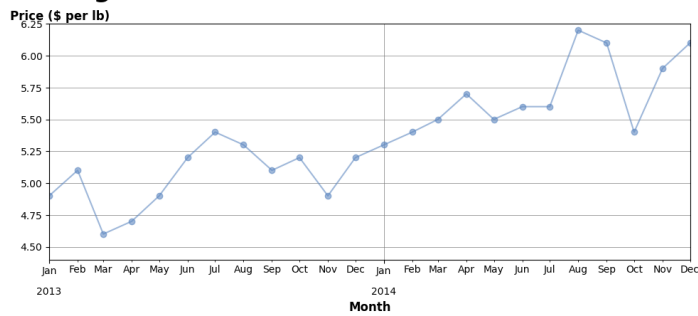
Average Coffee Bean Price from 2013 to 2014

Chart 4 - Univariate - Line plot

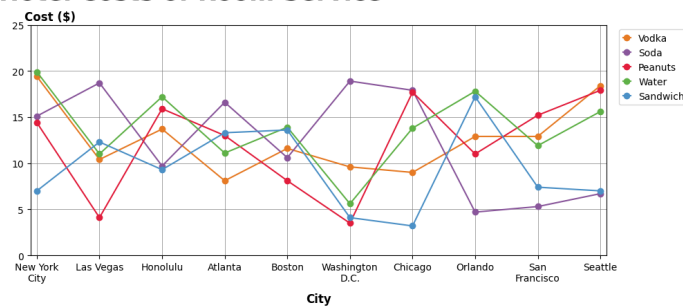
Hotel Costs of Room Service

Chart 5 - Multivariate - Line plot

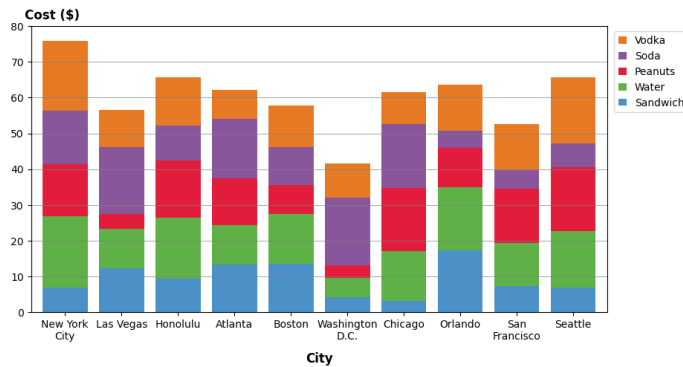
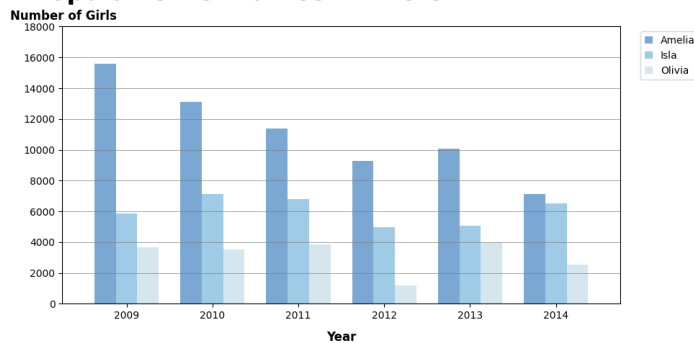
Hotel Costs of Room Service

Chart 5 - Multivariate - Stacked Bar plot

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Image

Popular Girls' Names in the UK



Chart

Chart 6 - Multivariate - Grouped Bar plot

Popular Girls' Names in the UK

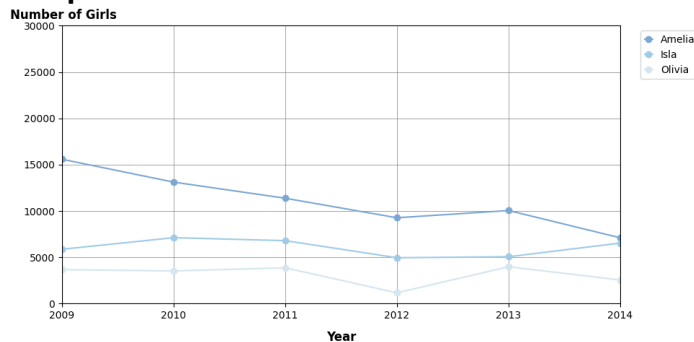


Chart 6 - Multivariate - Line plot

C Appendix: Questions

This appendix reports the full set of questions associated with each chart, grouped by chart number. For every chart, we list the underlying variable type (univariate or multivariate), the question type and subtype (e.g., *Retrieve Value*, *Find Extremum*, *Determine Range*, *Find Correlations/Trends*, *Make Comparisons*), the question text itself, and the multiple-choice options, with the correct answer shown in bold. Duplicate questions arising from multiple chart encodings of the same data (e.g., bar vs. line) have been collapsed to a single entry.

Chart 1

Variable	Question Type	Subtype	Question	Options
Univariate	Retrieve Value	Absolute value	What is the average internet speed in Japan?	1. 5.23 Mbps 2. 11.67 Mbps 3. 15.89 Mbps 4. 20.10 Mbps
Univariate	Find Extremum	Absolute value	In which country is the average internet speed the fastest in Asia?	1. South Korea 2. Malaysia 3. Sri Lanka 4. China
Univariate	Determine Range	Absolute value	What is the range of the average internet speed in Asia?	1. 0 - 18.71 Mbps 2. 13.15 - 17.82 Mbps 3. 9.62 - 18.71 Mbps 4. 10.24 - 14.16 Mbps
Univariate	Make Comparisons	Absolute value	How many countries in Asia is the average Internet speed slower than South Korea?	1. 6 countries 2. 7 countries 3. 8 countries 4. 9 countries

Chart 2

Variable	Question Type	Subtype	Question	Options
Univariate	Retrieve Value	Absolute value	What was the price of a barrel of oil in February 2015?	1. \$56.27 2. \$45.61 3. \$43.77 4. \$37.36
Univariate	Find Extremum	Absolute value	In which month was the price of a barrel of oil the lowest in 2015?	1. May 2. June 3. July 4. December
Univariate	Determine Range	Absolute value	What was the price range of a barrel of oil in 2015?	1. \$32.26 - \$66.49 2. \$43.77 - \$63.73 3. \$37.36 - \$59.88 4. \$37.36 - \$63.73
Univariate	Find Correlations/Trends	Absolute value	Over the course of the second half of 2015, was the price of a barrel of oil rising, falling or staying the same?	1. Rising 2. Falling 3. Staying
Univariate	Make Comparisons	Absolute value	About how much did the price of a barrel of oil rise from June to September in 2015?	1. \$3.57 2. \$48.91 3. \$21.04 4. \$26.37

Chart 3

Variable	Question Type	Subtype	Question	Options
Multivariate	Retrieve Value	Absolute value	What is the approval rating of Republicans among the people who have the education level of Postgraduate Study?	1. 37 2. 46 3. 54 4. 35
Multivariate	Find Extremum	Relative value	What is the education level of people in which the Democrats have the lowest approval rating?	1. High School Graduate or Less 2. Some College Degree 3. College Graduate 4. Postgraduate Study
Multivariate	Make Comparisons	Boolean	Is the approval rating of Republicans for the people who have the education level of Some College Degree is lower than that for the people who have the education level of Postgraduate Study.	1. TRUE 2. FALSE

Chart 4

Variable	Question Type	Subtype	Question	Options
Univariate	Retrieve Value	Absolute value	What was the average price of a pound of coffee beans in June 2013?	1. 5.2 2. 4.9 3. 5.4 4. 6.1
Univariate	Find Extremum	Absolute value	When was the average price of a pound of coffee beans at minimum?	1. April 2013 2. March 2013 3. June 2014 4. December 2014
Univariate	Determine Range	Absolute value	What was the range of the average price of a pound of coffee beans between January 2013 and December 2014?	1. \$4.3 - \$5.8 2. \$4.3 - \$5.9 3. \$4.6 - \$6.1 4. \$4.2 - \$5.9
Univariate	Find Correlations/Trends	Absolute value	Over the course of 2013, was the average price of a pound of coffee beans rising, falling or staying the same?	1. Rising 2. Falling 3. Staying
Univariate	Make Comparisons	Absolute value	For how many months was the average price of a pound of coffee beans the same as or cheaper than that in December 2013	1. 5 months 2. 6 months 3. 7 months 4. 8 months

Chart 5

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Variable	Question Type	Subtype	Question	Options
Multivariate	Retrieve Value	Absolute value	What is the cost of peanuts in Las Vegas?	1. 4.1 2. 11 3. 3.5 4. 8.1
Multivariate	Retrieve Value	Relative value	About what is the ratio of the cost of a sandwich to the total cost of room service in Seattle?	1. 01:10 2. 1:15 3. 2:10 4. 4:10
Multivariate	Find Extremum	Absolute value	In which city is the cost of soda the highest?	1. New York City 2. Washington D.C. 3. Atlanta 4. Las Vegas
Multivariate	Determine Range	Absolute value	What is the cost range of a sandwich in the cities?	1. 3.1-19.2 2. 4.1-13.6 3. 3.2-17.2 4. 7.4-13.6
Multivariate	Make Comparisons	Boolean	Is the ratio of the cost of peanuts to the cost of water in Las Vegas higher than that of San Francisco?	1. TRUE 2. FALSE
Multivariate	Make Comparisons	Boolean	Is the cost of water in Atlanta higher than that of Honolulu?	1. TRUE 2. FALSE

Chart 6

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Variable	Question Type	Subtype	Question	Options
Multivariate	Retrieve Value	Absolute value	What was the number of girls named 'Amelia' in 2010 in the UK?	1. 13116 2. 11376 3. 15585 4. 7118
Multivariate	Retrieve Value	Relative value	About what was the ratio of the number of girls named 'Amelia' to those named 'Isla' in 2014 in the UK?	1. 03:01 2. 4:1 3. 1:1 4. 5:1
Multivariate	Find Extremum	Absolute value	Over the course of years between 2009 and 2014, when was the number of girls named 'Amelia' at the maximum?	1. 2009 2. 2011 3. 2010 4. 2012
Multivariate	Find Correlations/Trends	Absolute value	Was the number of girls named 'Isla' rising, falling or staying the same from 2009 to 2012?	1. Rising 2. Falling 3. Staying
Multivariate	Make Comparisons	Boolean	In the UK, the number of girls named 'Amelia' in 2014 was more than it was in 2013.	1. TRUE 2. FALSE
Multivariate	Make Comparisons	Boolean	Over the course of years between 2009 and 2014, the number of girls named 'Isla' was always more than 'Olivia'.	1. TRUE 2. FALSE